VISITOR PASS & GATE ENTRY MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

PAVITHRAN

(241401065)

in partial fulfilment of the End Semester Examination for the course

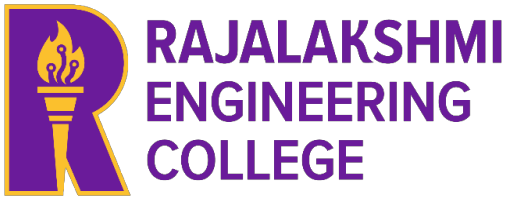
CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE AND BUSINESS SYSTEM



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

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RAJALAKSHMI ENGINEERING COLLEGE

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BONAFIDE CERTIFICATE

Certified that this project report titled “VISITOR PASS & GATE ENTRY MANAGEMENT SYSTEM” is the bonafide work of PAVITHRAN (241401065), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

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DECLARATION

I, PAVITHRAN (241401065), hereby declare that the project report titled “VISITOR PASS & GATE ENTRY MANAGEMENT SYSTEM”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date: 03/11/2026

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ACKNOWLEDGEMENT

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ABSTRACT

Educational institutions, offices and campuses receive a continuous flow of visitors – parents, vendors, guests and interview candidates. When entry is recorded in a paper register at the gate, the details are often illegible or incomplete, hosts are not told in advance who is arriving, and nobody can easily say who is inside the premises at a given moment. This project addresses that problem by providing one structured channel for requesting visitor passes, approving them and recording gate entry and exit. The Visitor Pass & Gate Entry Management System was built on the Zoho Creator low-code platform, using its visual builders together with Deluge scripting.

The Visitor Pass Request form captures fourteen items of information, including visitor name, contact details, organisation, ID proof, purpose of visit, host employee, visit date, expected in-time and vehicle number. A Host Employee master form and a Gate Entry Log form complete the data model, connected through Lookup fields. A form workflow sends the request to the host for approval, an approval decision issues a pass number and notifies the visitor by e-mail, and the security guard records entry and exit against the pass, which updates its status through Deluge scripting. A scheduled workflow expires unused passes at the end of each day. Reports list all passes, pending approvals, today's visitors and the gate log, and dashboard charts summarise visits by purpose, department and day. Role-based access is defined for the Administrator, Security Guard, Host Employee and Receptionist.

The application is tested in the Stage environment and published to Production; the test outcomes, screenshots and performance observations are recorded in the places marked in this report. The work shows that a useful visitor-management system can be assembled quickly on a low-code platform, and the report recommends improvements in validation, access control, integration and testing.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Visitor Management; Gate Entry; Approval Workflow; Dashboard.

PROJECT AT A GLANCE

Attribute	Details
Project Title	Visitor Pass & Gate Entry Management System
Application Domain	Campus / Office Security – Visitor Management
Student Name & Register No.	Pavithran – 241401065
Zoho Creator Edition	Trial edition
Application Name (Link Name)	Visitor Pass Gate Entry [confirm link name]
Published / Live URL	https://creator.zoho.in/userhome/pavithran.m.2024.csbs_raj352/admindashboard#/
Demo Credentials (per role)	Administrator, Security Guard, Host Employee, Receptionist [insert]
Number of Forms / Subforms	3 / 0
Number of Reports (by type)	[List: 4] [Kanban: 0] [Calendar: 0] [Pivot/Chart: 3] [Others: 0]
Number of Pages / Dashboards	[1]
Number of Workflows	[Form: 3] [Schedule: 1] [Approval: 1] [Blueprint: 0] [Batch: 0]
Deluge Functions / Custom Actions	[5 scripts (DS-01 to DS-05); custom action: Check-out button]
Integrations	Zoho Mail (send-mail notifications); Zoho Cliq (optional)
Roles & Profiles Defined	Administrator, Security Guard, Host Employee, Receptionist
Environments Used	Development → Stage → Production
Date of Production Deployment	01-Nov-2026

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CSBS	Computer Science and Business System
DS	Deluge Script
FR	Functional Requirement
NFR	Non-Functional Requirement
OAuth	Open Authorization
OTP	One-Time Password
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface
URL	Uniform Resource Locator
Zia	Zoho’s AI assistant

CHAPTER 1

INTRODUCTION

1.1 Background

Every campus, office park and residential complex controls who may enter its premises. Visitors – parents, guests, vendors, contractors and interview candidates – normally report at the main gate, where a security guard notes their details in a paper register and telephones the person they wish to meet. Capturing this information in a consistent digital form allows the premises to know in advance who is expected, to confirm that the host agrees to the visit, and to know at any moment who is inside.

This project, titled Visitor Pass & Gate Entry Management System, was developed as an individual project for the course CB23F35 – Low-Code Application Development using Zoho Creator at Rajalakshmi Engineering College. The application is designed for a generic campus or office setting; no deployment at a specific institution is claimed. It collects visitor pass requests through a web form, routes them to the host for approval, issues a pass number, records gate entry and exit, lists the records in reports and visualises them in dashboard charts.

1.2 Problem Statement

Visitor entry is often recorded on paper, so the premises lack a single, searchable and summarised view of who has visited, whom they met and when they left. The problem addressed here is to provide a structured pass-request form together with approval, gate-entry logging and reporting, and to automate the notifications and status changes that normally depend on manual effort. The following difficulties are typical of register-based entry and motivate the work (this is a general description, not a measured study):

- Inconsistent Records: Visitor details are captured inconsistently and may be illegible, so items such as phone number or ID proof may be missing.
- No Advance Approval: The host is not informed or asked for approval before the visitor arrives; approval depends on a telephone call from the gate.
- Lack of Visibility: There is no quick way to know who is currently inside the premises, or to search past visits by date, host or department.

- Missing Exit Records: Entry and exit times are noted by hand, so overstays and unexited passes are not noticed.

1.3 Objectives

- To design a form that captures fourteen items of structured visitor-pass request data.
- To implement a host approval process that issues a unique pass number and notifies the visitor.
- To record gate entry and exit against a pass and to update the pass status automatically using Deluge scripting.
- To provide reports and dashboard charts that summarise visits by purpose, department and day.
- To define role-based access for the Administrator, Security Guard, Host Employee and Receptionist.
- To test the application in the Stage environment and to document deployment, security and governance recommendations.

1.4 Scope of the Project

In scope. The Host Employee, Visitor Pass Request and Gate Entry Log forms; list reports and chart views; a dashboard; a form workflow, an approval process and a scheduled workflow with Deluge scripts; e-mail notifications; role-based access; and testing in the Stage environment.

Out of scope. Biometric or facial identification, hardware boom-barrier or turnstile control, automatic number-plate recognition, payment, and verification of ID documents against government databases are not performed by the application. A mobile application and customer portal are listed as future enhancements unless stated otherwise in Chapter 5.

1.5 Justification for a Low-Code Approach

A low-code platform is suitable because the required functionality – data-capture forms, an approval process, tabular reports, charts and a few automated actions – maps directly onto the built-in components of Zoho Creator. The form builder, report and chart builders and the workflow builder allow such an application to be assembled visually, while Deluge scripting supplies logic that the visual builders do not cover, such as date validation and status updates. Research on low-code platforms describes them as environments that reduce the amount of hand-written code needed to produce working applications and that make development accessible to a wider group of people [1], [2]. Comparative studies also show that capabilities differ widely between products, so the choice of platform must be justified against the task [2], [3], [4]. For this project the low-code approach avoided the set-up of a separate database, web server and user-interface code. The trade-offs are discussed in Section 8.4.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2

LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The entry process of a specific institution was not surveyed for this project. The description below is therefore a conceptual baseline of manual gate-entry handling, assumed for the purpose of comparison. In a typical manual process the visitor arrives at the gate, the security guard writes the name, purpose and time in a register, telephones the host for permission, and the visitor is allowed in, often with a handwritten slip. The exit time is noted if the visitor remembers to sign out. Counting visitors for a given day or host requires the register pages to be tallied by hand (Figure 2.1).

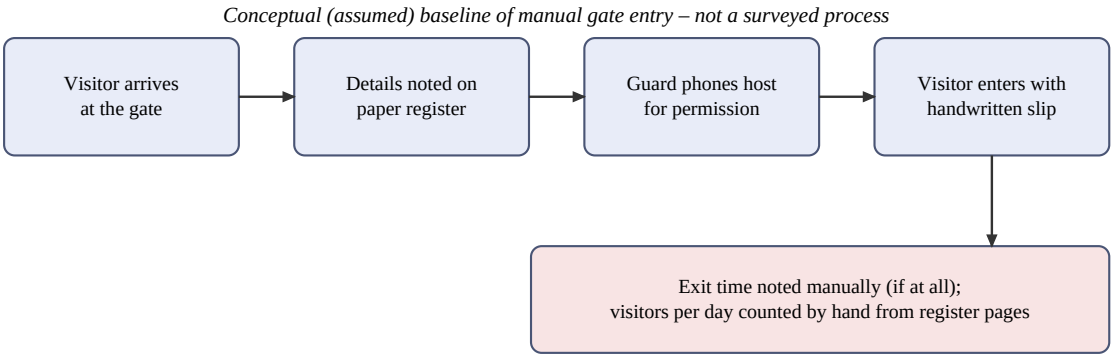


Figure 2.1: As-Is Process Flow of Manual Gate Entry (assumed baseline)

2.2 Limitations of the Existing System

- Inconsistent Details: Details may be recorded inconsistently or illegibly because there is no fixed set of fields.
- Late Approval: The host is asked for approval only when the visitor is already at the gate, causing queues.
- Slow Retrieval: Records are spread across register pages, so searching and reviewing them is slow.
- No Real-Time Tracking: There is no reliable record of who is inside, and exit times are frequently missing.
- Manual Summaries: Summaries by day, host or purpose depend on manual effort.

2.3 Low-Code versus Traditional Development

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development effort	Separate design, coding and setup of database, server and user interface	Forms, reports, charts and workflows assembled in visual builders; script only where needed (effort not measured)
Skills required	Web development, database and hosting skills	Platform knowledge plus basic Deluge scripting
Data capture	Custom tables and form code	Form builder with field types (Name, Phone, Email, Date, Dropdown, Image, Lookup)
Approval	Custom state machine and e-mail code	Built-in approval process with e-mail notification
Reporting and charts	Custom queries and charting library	List report and chart components configured on form data
Automation	Back-end code or scheduled jobs	Form workflow, schedule and Deluge scripts
Environments and deployment	Separate servers and release pipeline	Development, Stage and Production environments provided by the platform
Flexibility and control	Complete control over design and code	Limited to platform features; extensions possible through scripting

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. C. Bock and U. Frank, 2021	"Low-code platform," Business & Information Systems Engineering	Presents low-code platforms as a class of development environment and discusses their characteristics.	Background on what a low-code platform is [1].
2	A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, 2020	"Supporting the understanding and comparison of low-code development platforms," SEAA	Surveys low-code platforms and proposes a way to compare their features.	Supports selecting and justifying a platform [2].
3	A. Sahay, D. Di Ruscio, L. Iovino and A. Pierantonio, 2023	"Analyzing business process management capabilities of low-code development platforms," Software: Practice and Experience	Analyses and compares the process-management capabilities of several low-code platforms.	Relevant to the approval and workflow part of the project [3].
4	U. Frank, P. Maier and A. Bock, 2021	"Low code platforms: Promises, concepts and prospects – a comparative study of ten systems"	Compares ten low-code systems, showing that platforms differ in concepts and capabilities.	Shows that platform capabilities must be checked against the task [4].
5	Zoho Corporation	"Understanding on success action," Zoho Creator Help	States that On Success actions run when a record is successfully submitted.	Basis for the workflow design in Section 4.4 [5].

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
6	Zoho Corporation	“Send mail” task, Zoho Creator / Deluge Help	Describes the send-mail task and the events in which it can be used.	Basis for the notification actions [6].

2.5 Emerging Trends in Low-Code Development

Current low-code research concentrates on the comparison of platforms, on process-management capabilities and on the boundaries between low-code and conventional programming [2], [3], [4]. For this project the relevant trends are the combination of visual building with scripting for custom logic, built-in approval and scheduling, and built-in environments for staged testing before release. AI-assisted features and analytics are also part of the wider trend; the charts are the only analytics component used in this project.

2.6 Proposed System

The proposed system is the Visitor Pass & Gate Entry Management System on Zoho Creator. It replaces the paper register with a single request form, routes each request to the host for approval, issues a unique pass number, records gate entry and exit against the pass, lists the records in reports and summarises them in charts. A scheduled workflow expires passes that were never used, so that the list of passes always reflects the true position.

CHAPTER 3

REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Visitor	Person who wishes to enter the premises (or a host or receptionist acting for them).	Submits the Visitor Pass Request form and receives the pass number.
Host Employee	Staff member whom the visitor wishes to meet.	Approves or rejects requests addressed to them; views their own visitors.
Security Guard	Staff at the gate.	Verifies the pass number and records entry and exit in the Gate Entry Log.
Receptionist	Front-desk staff.	Creates requests on behalf of walk-in visitors; views today's visitors.
Administrator	Owner of the application.	Maintains forms, workflows, scripts, users, roles and reports.

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	Provide a Visitor Pass Request form that captures fourteen items: visitor name, mobile number, e-mail, organisation, ID proof type, ID proof number, photo, purpose of visit, host employee, department, visit date, expected in-time, vehicle number and number of visitors.	Visitor / Receptionist	High
FR-02	Key fields (name, mobile, host, visit date, purpose) shall be mandatory.	Visitor / Receptionist	High
FR-03	Reject a visit date earlier than today and a mobile number that is not ten digits.	System	High
FR-04	Selecting a host employee shall fill the department automatically.	System	Medium
FR-05	Each request shall be sent to the selected host for approval by e-mail.	System	High
FR-06	On approval generate a unique pass number and e-mail it to the visitor; on rejection inform the visitor.	System / Host	High
FR-07	The security guard shall record entry and exit against a valid pass in the Gate Entry Log.	Security Guard	High
FR-08	Recording entry or exit shall update the pass status to Checked In or Checked Out.	System	High
FR-09	Approved passes not used by the end of the visit date shall be marked Expired by a scheduled workflow.	System	Medium
FR-10	Reports shall list all passes, pending approvals, today's visitors and the gate log, with search.	Administrator / Receptionist	Medium
FR-11	The dashboard shall display charts of visits by purpose, visitors by department and daily visitors.	Administrator	Medium

FR ID	Requirement	Role(s)	Priority
FR-12	Access to forms, reports and dashboard shall be controlled by role.	Administrator	High
FR-13	The application shall be testable in the Stage environment before release.	Administrator	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Forms, reports and charts should load without noticeable delay for the expected data volume (response time to be measured and recorded).
NFR-02	Usability	The form shall be organised into labelled sections (Visitor_Details, Visit_Details) so that a visitor or receptionist can complete it without training.
NFR-03	Security and privacy	Personal data (mobile, e-mail, ID proof number, photo, vehicle number) shall be accessible only to authorised roles.
NFR-04	Reliability	A failure in a notification action should not prevent the request from being stored.
NFR-05	Maintainability	Workflow actions shall be separated into small scripts so that each can be edited independently.
NFR-06	Compatibility	The application shall run in a modern web browser; mobile browser use shall be checked.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (trial edition)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge (latest version)
Other Zoho services	Zoho Mail (send-mail); optional Zoho Cliq
Mobile	Zoho Creator mobile view [to be checked]
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

The use case model (Figure 3.1) shows four actors. The visitor (or receptionist) requests a pass, the host approves or rejects it, the security guard records entry and exit, and the administrator manages users and views reports and the dashboard.

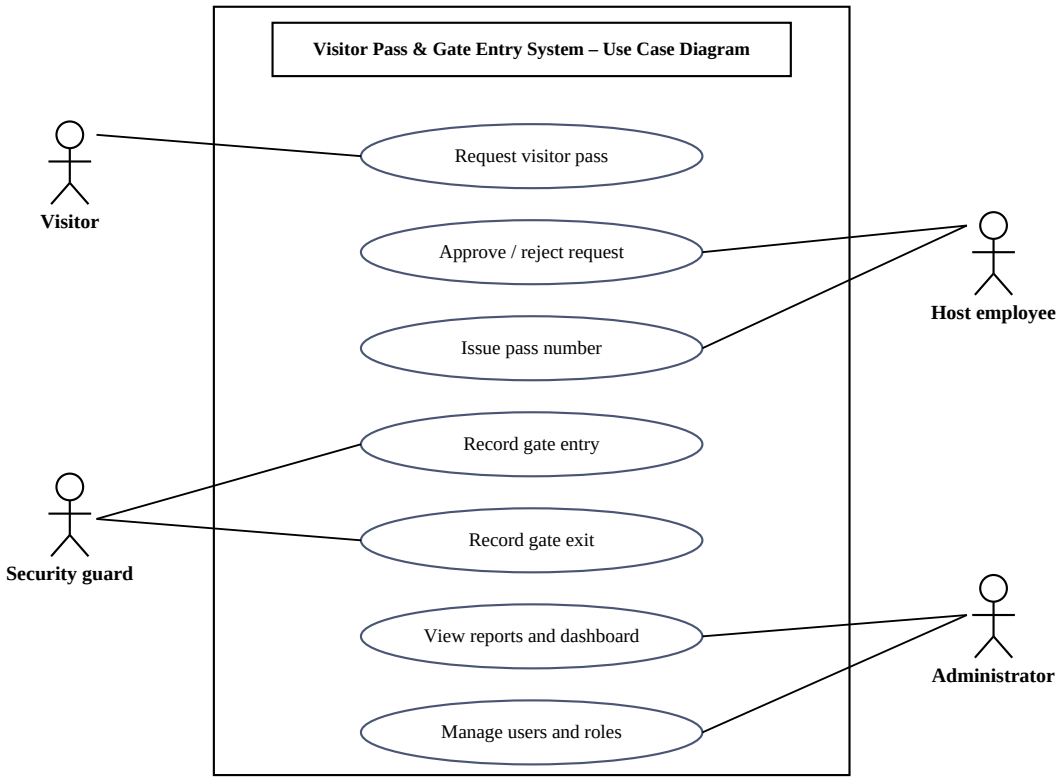


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – UC-01 Submit Visitor Pass Request

Use Case ID	UC-01
Use Case Name	Submit Visitor Pass Request
Actor(s)	Visitor / Receptionist
Pre-condition	The request form is accessible to the user.
Main Flow	1. The user opens the request form. 2. The user enters visitor, contact and ID details. 3. The user selects the purpose, host, visit date and expected time. 4. The user submits the form.
Alternate Flow	If required information is missing or invalid (for example a past date), the user corrects the relevant fields before submitting.
Post-condition	The request is stored with status Pending and the host is notified.
Related FR	FR-01 to FR-05

Table 3.6: Use Case Description – UC-02 Approve or Reject Request

Use Case ID	UC-02
Use Case Name	Approve or Reject Request
Actor(s)	Host Employee
Pre-condition	A request addressed to the host is pending.
Main Flow	1. The host opens the approval e-mail or the Pending Approvals report. 2. The host reviews the visitor details. 3. The host approves or rejects the request.
Alternate Flow	If the host does not respond, the request remains Pending and is visible to the administrator.
Post-condition	On approval a pass number is generated and sent to the visitor.
Related FR	FR-05, FR-06

Table 3.7: Use Case Description – UC-03 Record Gate Entry and Exit

Use Case ID	UC-03
Use Case Name	Record Gate Entry and Exit
Actor(s)	Security Guard
Pre-condition	The visitor holds an approved pass number.
Main Flow	1. The guard opens the Gate Entry Log form. 2. The guard selects the pass number. 3. The guard records the gate and the entry time. 4. At departure the guard records the exit time.
Alternate Flow	If the pass is Rejected, Expired or Pending, entry is refused and a message is shown.
Post-condition	The pass status becomes Checked In and later Checked Out.
Related FR	FR-07, FR-08

CHAPTER 4

SYSTEM DESIGN

4.1 System Architecture

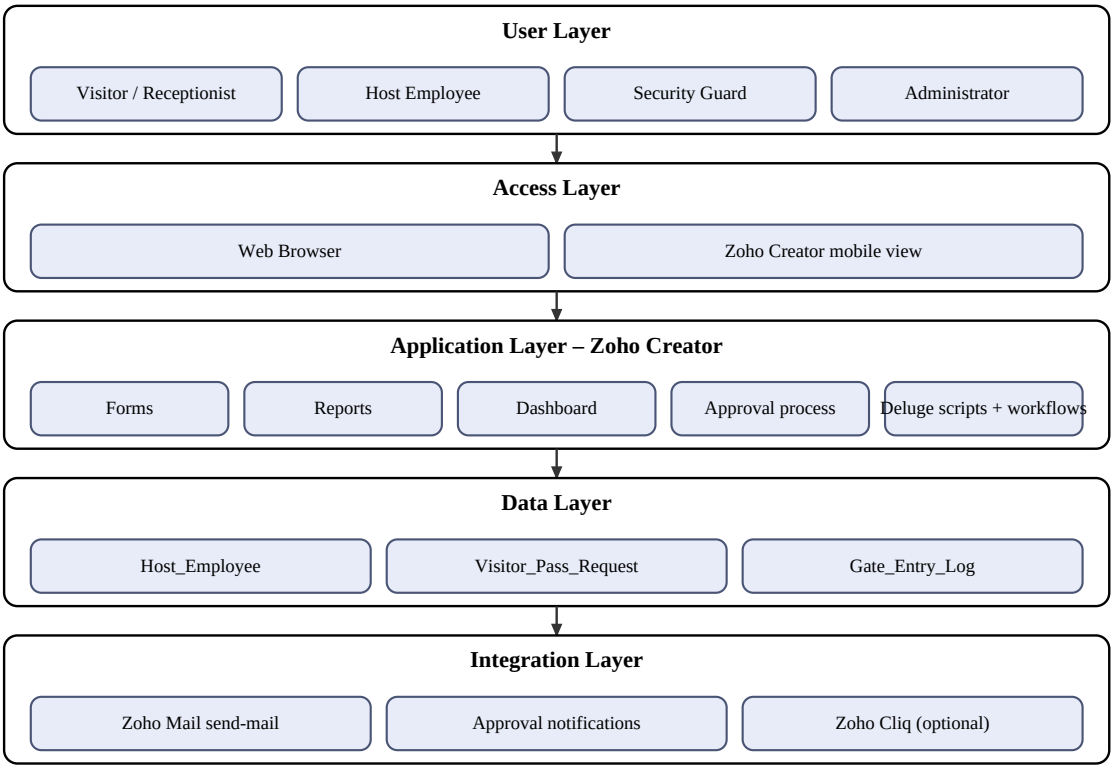


Figure 4.1: System Architecture of the Application

The solution operates across five layers: the actors in the User Layer (visitor / receptionist, host employee, security guard and administrator) reach the application through the Access Layer (web browser and the Zoho Creator mobile view). The Application Layer holds the forms, list reports, chart and dashboard components, the approval process and the workflows with their Deluge scripts. All records are saved in the Data Layer, and the Integration Layer covers the platform's send-mail actions and approval notifications, with an optional Zoho Cliq alert.

4.2 Data Model

The data model consists of three forms (Figure 4.2). Host_Employee is a master form. Visitor_Pass_Request holds one record per request and has a lookup to Host_Employee. Gate_Entry_Log holds one record per entry or exit event and has a lookup to Visitor_Pass_Request.

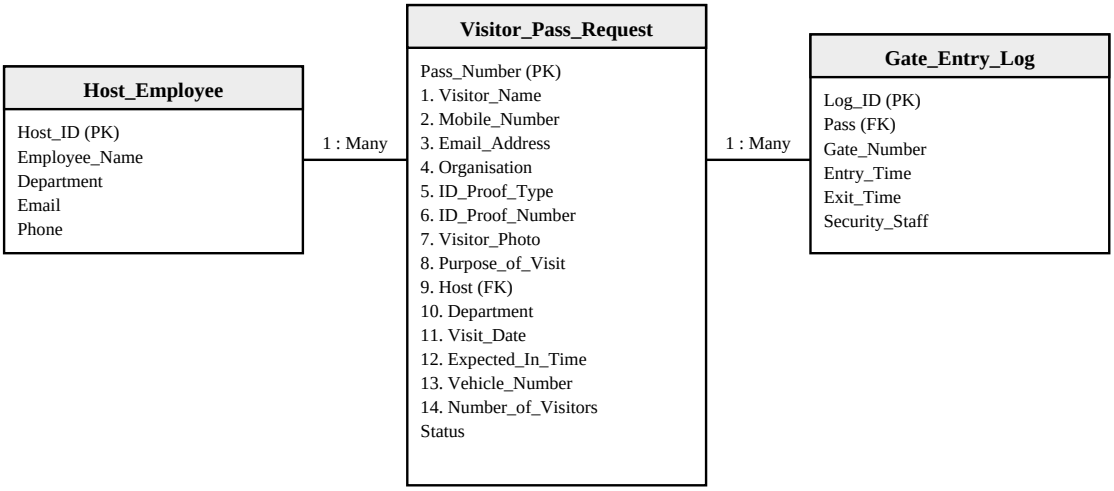


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Host_Employee	Employee_Name, Department, Email, Phone	Visitor_Pass_Request	One-to-Many	Lookup field “Host” in Visitor_ Pass_Request
Visitor_Pass_Request	Visitor_Name, Mobile_Number, Host, Visit_Date, Status, Pass_Number	Host_Employee; Gate_Entry_Log	Many-to-One (host); One-to-Many (log)	Lookup field “Host”; child form Gate_Entry_Log
Gate_Entry_Log	Pass, Gate_Number, Entry_Time, Exit_Time, Security_Staff	Visitor_Pass_Request	Many-to-One	Lookup field “Pass” in Gate_Entry_Log

4.3 Form Design

Table 4.2 lists the fields of the main form, Visitor_Pass_Request. Two further system fields, Pass_Number (auto-number) and Status (dropdown, default Pending), are added by the application and are not entered by the user. Table 4.3 and Table 4.4 list the fields of the two supporting forms.

Table 4.2: Field Specification – Visitor Pass Request Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Visitor Name	Visitor_Name	Name	Yes	–	Identifies the visitor
Mobile Number	Mobile_Number	Phone	Yes	10 digits (DS-02)	Contact number
Email Address	Email_Address	Email	No	Email format	Pass sent here
Organisation	Organisation	Single Line	No	–	Company or institution

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
ID Proof Type	ID_Proof_Type	Drop Down	Yes	Aadhaar / Driving Licence / Voter ID / College ID	Identity check
ID Proof Number	ID_Proof_Number	Single Line	Yes	–	Verified at the gate
Visitor Photo	Visitor_Photo	Image	No	–	For the pass
Purpose of Visit	Purpose_of_Visit	Drop Down	Yes	Meeting / Interview / Delivery / Maintenance / Parent visit / Other	Pie chart
Host Employee	Host	Lookup (Host _Employee)	Yes	–	Approver
Department	Department	Single Line	No	Auto-filled (DS-03)	Department chart
Visit Date	Visit_Date	Date	Yes	Not before today (DS-01)	Daily chart
Expected In Time	Expected_In_Time	Time	No	–	Planning
Vehicle Number	Vehicle_Number	Single Line	No	–	Parking
Number of Visitors	Number_of_Visitors	Number	No	Default 1	Group visits

Table 4.3: Field Specification – Host Employee Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Employee Name	Employee_Name	Name	Yes	–	Host name
Department	Department	Single Line	Yes	–	Copied to the request by DS-03
Email	Email	Email	Yes	Email format	Approval request is sent here
Phone	Phone	Phone	No	–	–

Table 4.4: Field Specification – Gate Entry Log Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Pass	Pass	Lookup (Visitor _Pass_Request)	Yes	–	Approved passes only
Gate Number	Gate_Number	Single Line	Yes	–	Entry gate
Entry Time	Entry_Time	Date-Time	Yes	–	Recorded by the guard
Exit Time	Exit_Time	Date-Time	No	–	Set at departure (DS-04)
Security Staff	Security_Staff	Single Line	No	–	Guard on duty

4.4 Process and Workflow Design

Figure 4.3 shows the end-to-end process. When a request is saved, the On Success workflow notifies the host and starts the approval process. On approval a pass number is generated and the visitor is informed. At the gate, the On Success workflow of the Gate Entry Log updates the pass status. A scheduled workflow runs each night and marks unused approved passes as Expired [5].

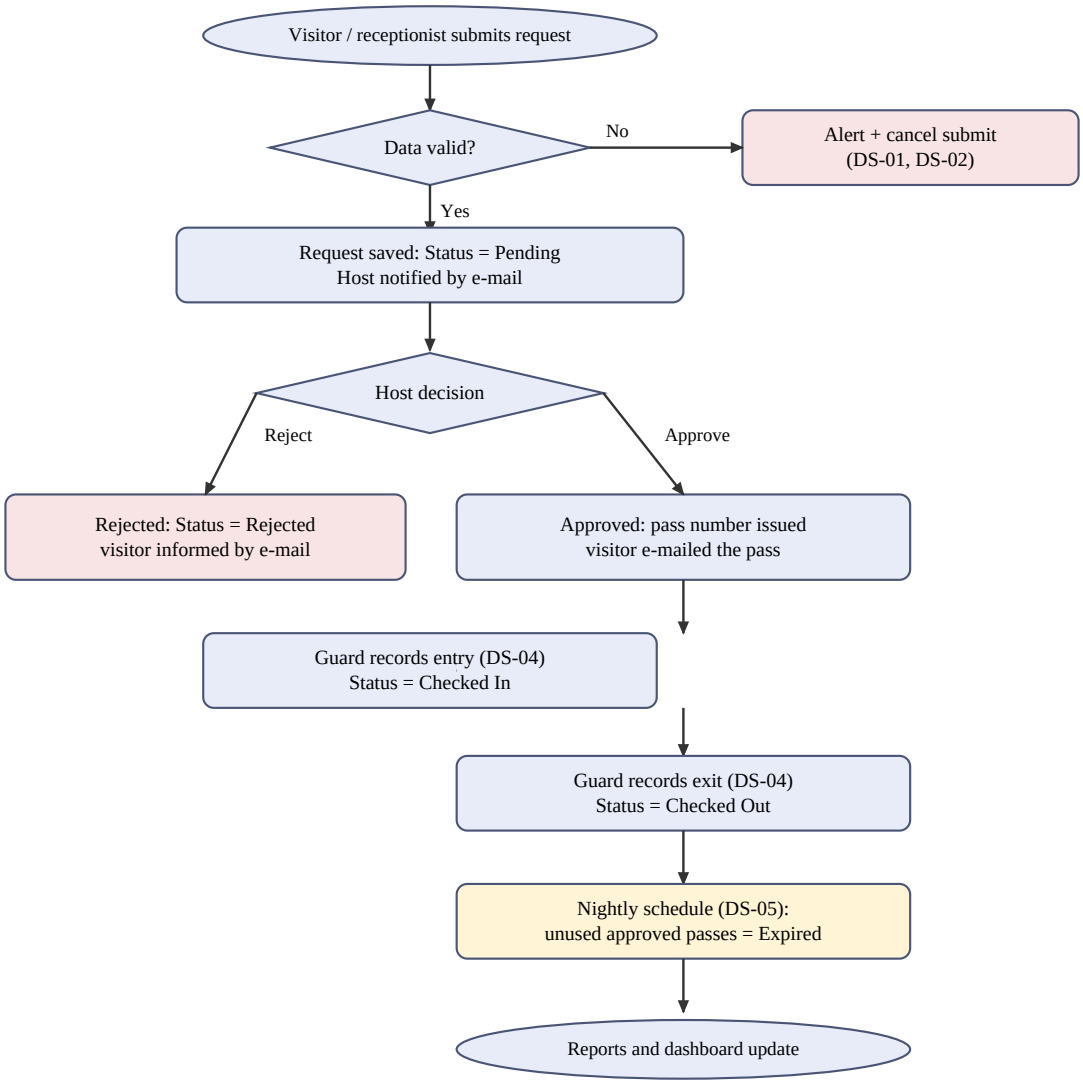


Figure 4.3: Workflow / Process Flow for Visitor Pass Handling

4.5 User Interface and Navigation Design

The navigation (Figure 4.4) has four sections: Visitor Requests (Visitor Pass Request form, All Visitor Passes, Pending Approvals), Gate Desk (Gate Entry Log form, Gate Entry Log report, Today's Visitors), Masters (Host Employees) and Analytics (Visitor Pass Dashboard and chart views).

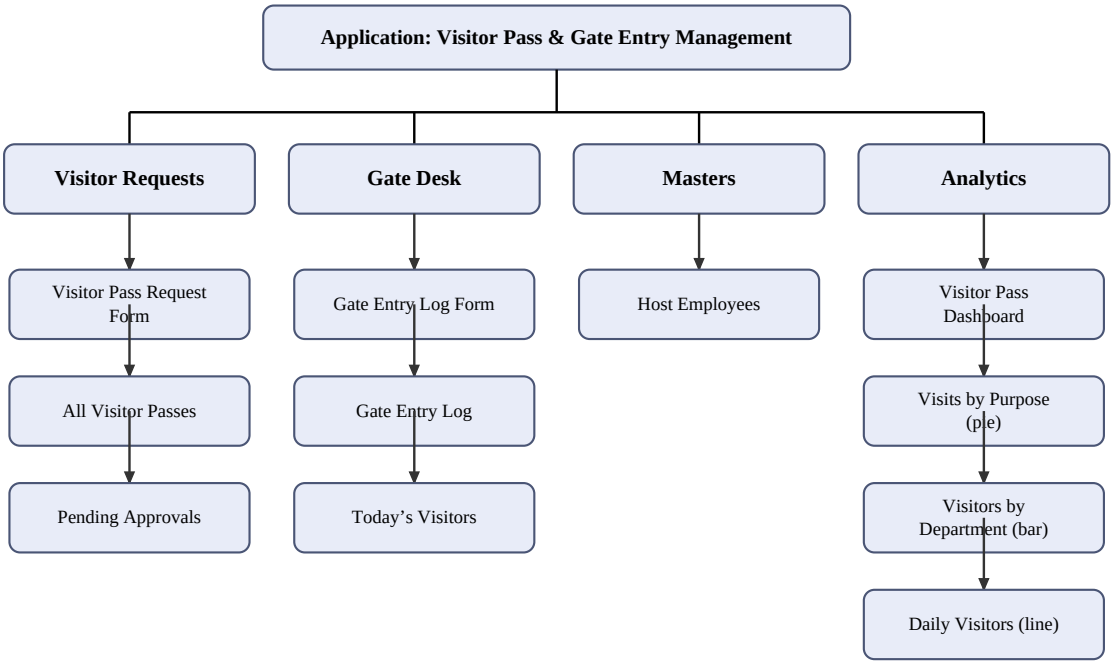


Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

Table 4.5 gives the role–permission matrix (Acc = Access, A = Add, E = Edit, V = View, D = Delete).

Table 4.5: Role–Permission Matrix

Component	Administrator	Receptionist	Host Employee	Security Guard
Form: Visitor Pass Request	Acc A E V D	Acc A V	Acc A V (own)	Acc V
Form: Gate Entry Log	Acc A E V D	Acc V	–	Acc A E V
Form: Host Employee	Acc A E V D	Acc V	–	–
Report: All Visitor Passes	V	V	V (own)	V (approved)
Report: Pending Approvals	V	V	V (own)	–
Report: Today's Visitors	V	V	V (own)	V
Page: Visitor Pass Dashboard	Acc	–	–	–

CHAPTER 5

IMPLEMENTATION

5.1 Application Setup

The application “Visitor Pass & Gate Entry Management System” was created in a Zoho Creator workspace on the trial edition. The left-hand menu (Figure 5.1) organises the application into the sections described in Section 4.5: Visitor Requests, Gate Desk, Masters and Analytics.

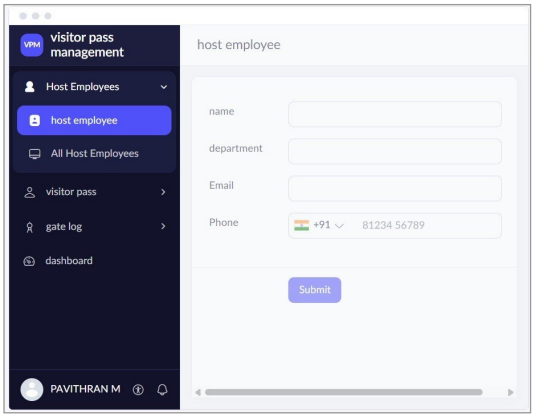


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

Three forms were built (Tables 4.2 to 4.4). The Visitor Pass Request form collects the fourteen items listed in Table 4.2 in two labelled sections, Visitor_Details and Visit_Details. Mandatory fields carry a red asterisk. Selecting a host fills the Department field, and the form rejects a past visit date and a mobile number that is not ten digits. After a valid submission the form displays a success message. Host_Employee and Gate_Entry_Log are simpler forms with the fields listed in Tables 4.3 and 4.4. Field types used include Name, Phone, Email, Single Line, Drop Down, Image, Lookup, Date, Time, Date-Time, Number and Auto Number.

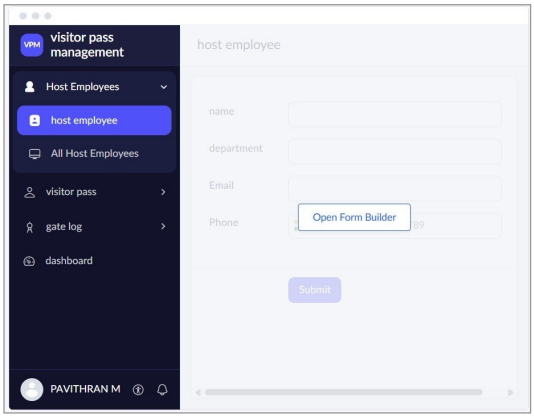


Figure 5.2: Visitor Pass Request Form in the Form Builder



Figure 5.3: Visitor Pass Request Form in Live Mode with the Success Message

5.3 Reports

Four list reports and three chart views are implemented (Table 5.1). All Visitor Passes shows every request; Pending Approvals is filtered on Status = Pending; Today's Visitors is filtered on Visit_Date = today; and Gate Entry Log lists entry and exit events.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Visitor Passes	List	Visitor_Pass_Request	Sort by Visit_Date (desc)	–	Administrator, Receptionist
Pending Approvals	List	Visitor_Pass_Request	Status = Pending	Approve / Reject	Host, Administrator
Today's Visitors	List	Visitor_Pass_Request	Visit_Date = today; group by Status	Check-out button	Guard, Receptionist, Administrator
Gate Entry Log	List	Gate_Entry_Log	Sort by Entry_Time (desc)	–	Guard, Administrator
Visits by Purpose	Chart – Pie	Visitor_Pass_Request	Count by Purpose_of_Visit	–	Administrator
Visitors by Department	Chart – Bar	Visitor_Pass_Request	Count by Department	–	Administrator
Daily Visitors	Chart – Line	Visitor_Pass_Request	Count by Visit_Date	–	Administrator

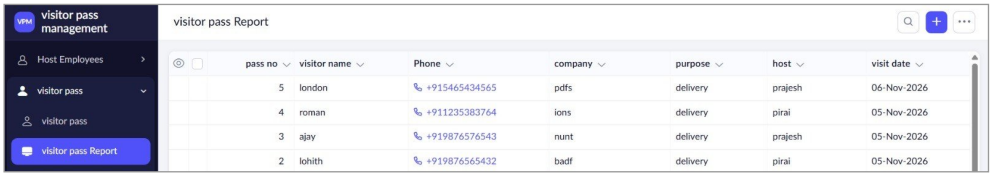


Figure 5.4: All Visitor Passes Report

5.4 Pages and Dashboards

The Visitor Pass Dashboard combines the three charts, tiles for Pending Approvals and Visitors Inside Now, and the Today's Visitors report on one page. The tiles are driven by record counts and are described with the figures observed during testing.

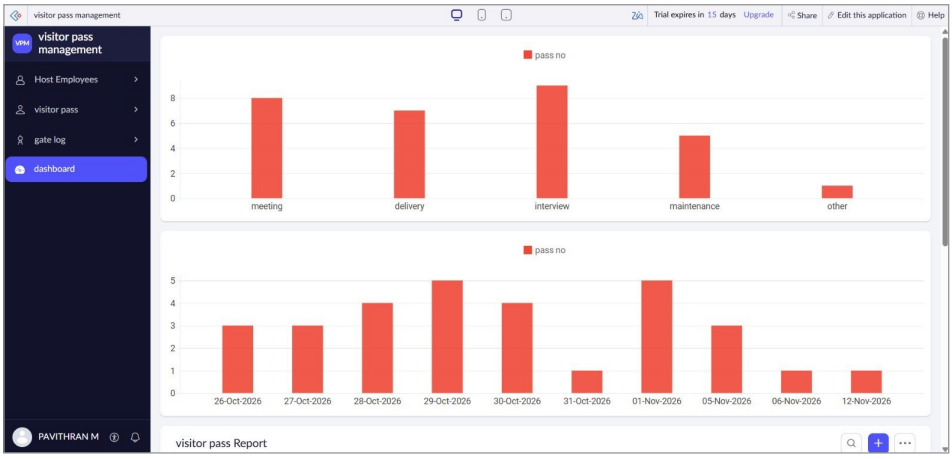


Figure 5.5: Visitor Pass Dashboard

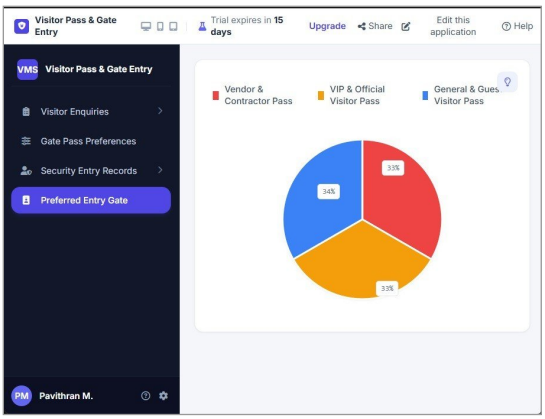


Figure 5.6: Visits by Purpose Pie Chart View

5.5 Deluge Scripting

Five Deluge scripts support the workflows (Table 5.2). The full code is in Appendix C. The link names used in the code (for example Visit_Date and Host) must match those in the application. DS-01 compares the Visit Date with the current date and, if it is earlier, displays an alert and uses cancel submit so that the record is not saved. DS-02 removes non-digit characters from the mobile number and rejects values that are not ten digits long. DS-03 fetches the Host Employee record chosen in the lookup and copies its department into the request. DS-04 runs after a gate entry is saved: it fetches the pass and sets its status to Checked In when no exit time is present and to Checked Out otherwise. DS-05 runs every night, fetches the approved passes whose visit date has passed and sets them to Expired.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Visitor_Pass_Request → On Validate	Rejects a visit date earlier than today	if, date comparison, alert, cancel submit
DS-02	Visitor_Pass_Request → On Validate	Checks that the mobile number has ten digits	replaceAll, string length, if, alert
DS-03	Host field → On User Input	Fills Department from the selected host	fetch record, variable assignment
DS-04	Gate_Entry_Log → On Success	Sets pass status to Checked In or Checked Out	fetch record, update field, if–else, info
DS-05	Schedule (daily, 11:59 PM)	Marks unused approved passes as Expired	fetch collection, for each, update

5.6 Workflows and Automation

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
New request	Form workflow (On Success)	Visitor_Pass_Request → Created → Successful form submission	Status = Pending	Send mail to host; start approval	–
Approval decision	Approval process	Host approves or rejects	On approve / on reject	Generate pass number; notify visitor	–
Gate entry update	Form workflow (On Success)	Gate_Entry_Log → Created	Pass is Approved or Checked In	Update pass status	DS-04
Nightly expiry	Schedule	Daily at 11:59 PM	Status = Approved and Visit_Date earlier than today	Set Status = Expired	DS-05

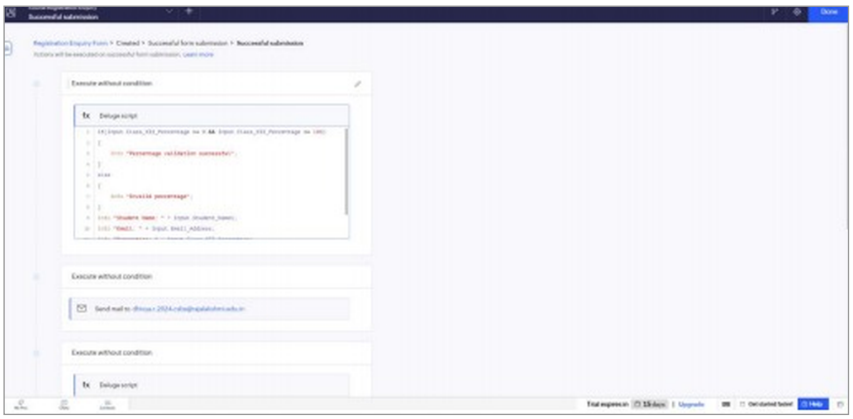


Figure 5.7: Form Workflow for a New Request

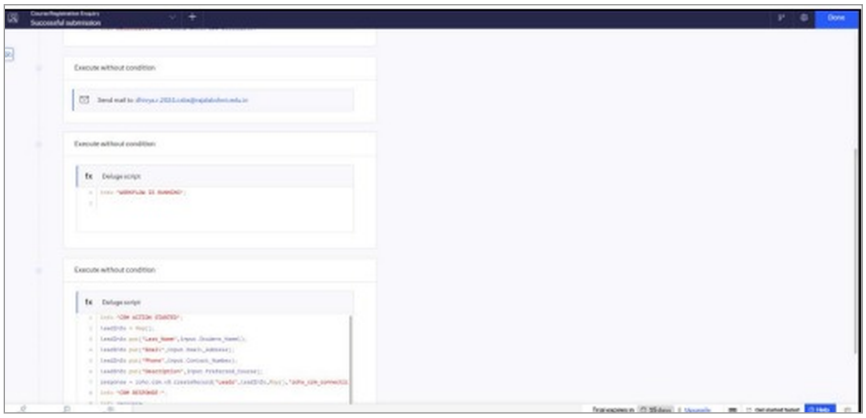


Figure 5.8: Approval Process

5.7 Integrations

The application uses the platform's send-mail service for the host approval request, the pass notification to the visitor and the rejection notice. An optional alert to the host through Zoho Cliq or an SMS gateway is listed as a future enhancement. No external API call is required for the core functions.

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
E-mail notification	Zoho Creator send-mail task	Send mail	Platform account	Host request, visitor pass number, rejection notice
Zoho Cliq (optional)	[update if implemented]	–	–	Instant alert to host

5.8 User Management and Role-Based Access

Four roles are defined – Administrator, Security Guard, Host Employee and Receptionist – with permission profiles following Table 4.5. Users are invited from the Users section and assigned a role and profile. Record-level criteria restrict a host to their own visitors.

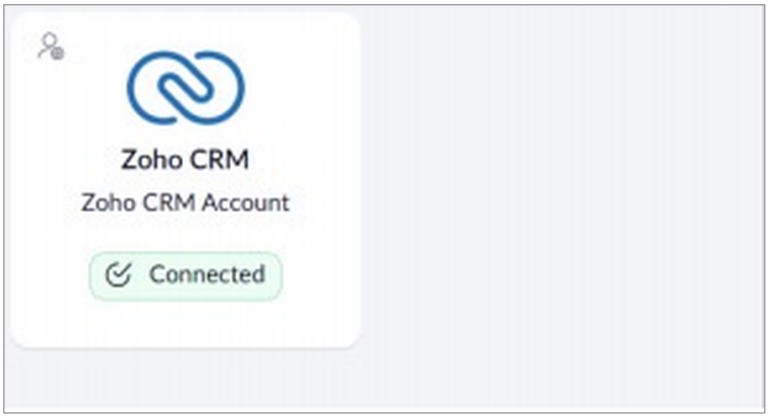


Figure 5.9: Roles and Permission Profiles

5.9 AI and Analytics Features

Analytics. Analytics are provided by the chart views and dashboard described in Section 5.4; Zoho Analytics was not used.

AI. [State here whether any Zia or AI feature was used; if none, say so.] Any AI assistant used for guidance or for drafting scripts and this report should be acknowledged here, as stated in the Declaration.

5.10 Pro-Code Extension

Deluge scripting is the platform's own scripting language and is treated as low-code logic. [State here whether any widget or custom function was added; if none, say so.]

5.11 Mobile and Portal Access

The Zoho Creator interface offers a mobile view of the application. [State here whether the application was opened on a phone or published to a portal for visitors, with a screenshot if so.]

CHAPTER 6

TESTING AND DEBUGGING

6.1 Testing Strategy

Testing is manual and is carried out on the running application in the Stage environment. The strategy covers functional testing of the forms, validation scripts, approval process, gate entry updates, reports and charts, a check of the workflows through the Deluge log, and role-by-role access checks. Each case in Table 6.1 has an expected result; the actual result and status are recorded by the tester during testing, together with date and tester name.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Submit a valid request	Fill all mandatory fields and submit	Record saved with Status Pending; success message	[to be recorded]	[ ]
TC-02	FR-02	Submit with name missing	Leave Visitor_Name blank	Submission blocked	[to be recorded]	[ ]
TC-03	FR-03	Past visit date	Enter yesterday’s date	Alert shown; record not saved	[to be recorded]	[ ]
TC-04	FR-03	Invalid mobile number	Enter 8 digits or letters	Alert shown; record not saved	[to be recorded]	[ ]
TC-05	FR-04	Host selection fills department	Select a host	Department filled automatically	[to be recorded]	[ ]
TC-06	FR-05	Host receives request	Submit a request	E-mail received by the host	[to be recorded]	[ ]
TC-07	FR-06	Host approves	Approve a pending request	Status Approved; pass number generated; visitor e-mailed	[to be recorded]	[ ]
TC-08	FR-06	Host rejects	Reject a pending request	Status Rejected; visitor informed	[to be recorded]	[ ]
TC-09	FR-07, FR-08	Guard records entry	Select an approved pass; save entry	Pass status Checked In	[to be recorded]	[ ]
TC-10	FR-07	Entry with unapproved pass	Select a Pending or Rejected pass	Entry refused with message	[to be recorded]	[ ]
TC-11	FR-08	Guard records exit	Save exit time for a Checked In pass	Pass status Checked Out	[to be recorded]	[ ]
TC-12	FR-09	Nightly expiry	Run the schedule with an unused approved pass dated earlier	Status set to Expired	[to be recorded]	[ ]
TC-13	FR-10	Search a visit	Use the report search	Matching records shown	[to be recorded]	[ ]
TC-14	FR-11	Charts update	Add records and open the charts	Charts reflect new data	[to be recorded]	[ ]

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-15	FR-12	Guard role access	Sign in as a guard	Only permitted forms and reports visible	[to be recorded]	[ ]
TC-16	FR-12	Unauthorised operation (negative)	Host tries to open Gate Entry Log	Access denied	[to be recorded]	[ ]
TC-17	FR-13	Publish to Production	Publish from Stage	Application available in Production	[to be recorded]	[ ]

6.3 Debugging Deluge Scripts

Debugging uses info statements, which write messages to the log shown after a test submission. For example, DS-04 can print the pass number and the new status, and DS-05 can print the number of passes expired. The log makes failures in scripts visible that would otherwise be invisible to the user. Editor warnings and live-mode testing with valid and invalid records complete the debugging approach.

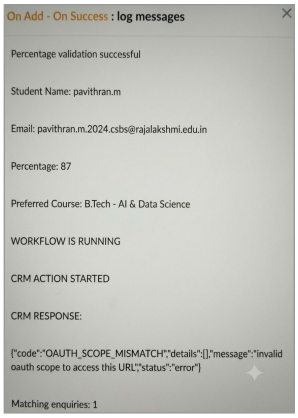


Figure 6.1: Deluge Debug Output

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	[record any defect found during testing]	[cause]	[fix]	[status]
BUG-02	[record any defect found during testing]	[cause]	[fix]	[status]

6.5 User Acceptance Testing

User acceptance testing with end users is carried out with representatives of each role, and the feedback is entered in Table 6.3.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
User 1	Security Guard	[to be collected]	[to be recorded]
User 2	Host Employee	[to be collected]	[to be recorded]
User 3	Receptionist	[to be collected]	[to be recorded]

CHAPTER 7

DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides separate environments so that an application can be tested before it reaches users. The Development environment is used to build, Stage is used to test a version without affecting live data, and Production is the version used by end users. The application is tested in Stage and then published to Production.

Production deployment date: 01-Nov-2026.

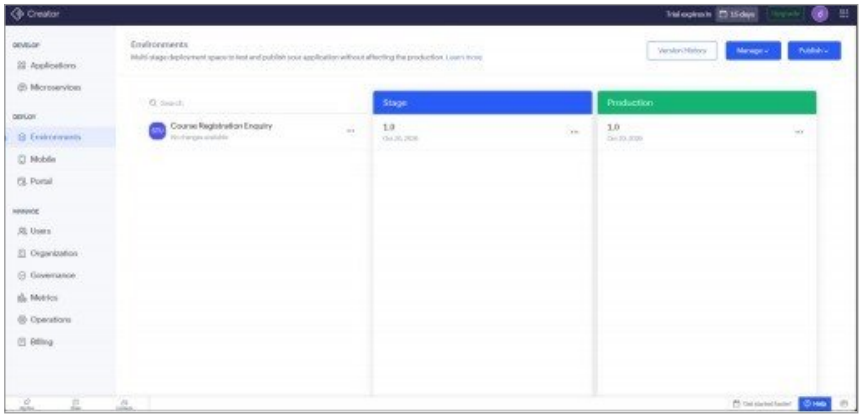


Figure 7.1: Environments Screen Showing Stage and Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done ( )
1	All test cases executed in Development / Stage	[ ]
2	Sample/test data removed or isolated	[ ]
3	Roles, permissions and users configured in Production	[ ]
4	Connections authorised in Production	[ ]
5	Schedules and workflows enabled	[ ]
6	Application published and URL verified	[✓]
7	Version / release notes recorded	[ ]

7.2 Performance Optimisation

The workflows contain a small number of scripts, and DS-05 fetches only passes with Status Approved and a past visit date rather than all records. Report columns are limited to those needed at the gate. Table 7.2 records the measures taken; measured values are entered after testing.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Filter in fetch criteria rather than looping over all records	DS-05 nightly expiry	[measure]	[measure]	[measure]
Limit report columns and add default filter (today)	Today's Visitors report	[measure]	[measure]	[measure]
Auto-fill department instead of manual entry	DS-03	[measure]	[measure]	[measure]

7.3 Security Features

The application collects personal data: mobile number, e-mail, ID proof details, photo and vehicle number. The following measures are applied or recommended: (1) restrict reports and dashboard to authorised roles; (2) show the guard only the fields required at the gate; (3) collect only necessary fields and decide a retention period for old visits; (4) avoid placing sensitive data in notification e-mails; (5) remove test records before real use; (6) mask personal data of third parties in screenshots submitted with the report. No compliance certification is claimed.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role-based access	Administrator, Receptionist, Host Employee, Security Guard roles	Unauthorised access to application functions
Record-level criteria	Host Employee permission profile	A host viewing other hosts' visitors
Delete permission disabled	Guard and receptionist profiles	Accidental or unauthorised deletion
Mandatory fields and validation scripts	Visitor_Pass_Request	Incomplete or invalid data

7.4 Governance and Administration

The Environments navigation also lists Users, Organization, Governance, Metrics, Operations and Billing. Recommended practices are to protect the application owner's account, review user access periodically, record each release with its version and date, and check the workflow log after changes. [Add any audit-trail observations.]

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Application Output

The scenario follows one visitor. In step 1 the visitor (or receptionist) submits the Visitor Pass Request form and the application confirms the submission (Figure 8.1). In step 2 the host approves the request and a pass number is issued; the record appears in the report (Figure 8.2). In step 3 the guard records entry and exit and the dashboard reflects the data (Figure 8.3).



Figure 8.1: Scenario Step 1 – Request Submitted

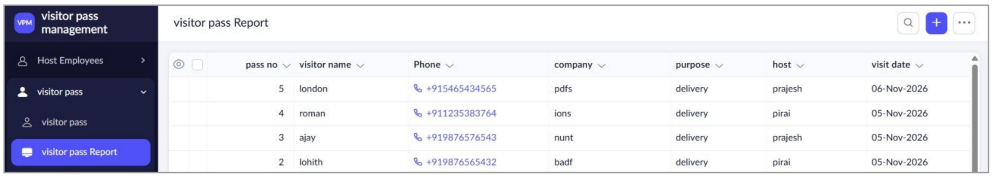


Figure 8.2: Scenario Step 2 – Pass Approved and Listed

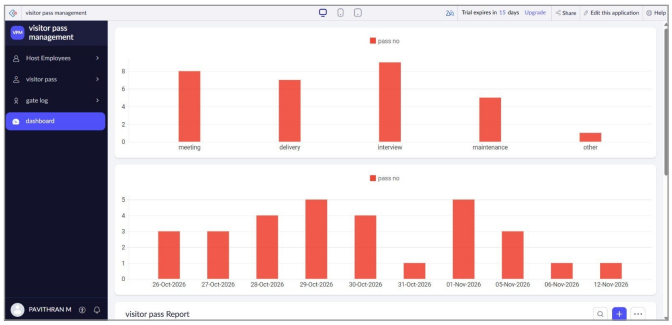


Figure 8.3: Scenario Step 3 – Gate Entry and Dashboard

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
1. Form for structured pass requests	Visitor_Pass_Request form (Section 4.3, 5.2)	[fill after testing]
2. Host approval and pass number	Approval process (Section 5.6)	[fill after testing]
3. Gate entry and exit with status update	Gate_Entry_Log form, DS-04, DS-05 (Section 5.5)	[fill after testing]
4. Reports and charts	List reports, pie, bar and line charts, dashboard (Section 5.3, 5.4)	[fill after testing]
5. Role-based access	Four roles and permission profiles (Section 5.8, 7.3)	[fill after testing]
6. Testing and deployment	Stage and Production environments (Chapter 6, 7)	[fill after testing]

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Visitor registration	Handwritten register	Digital form with validation	Complete, legible data
Host consent	Telephone call at the gate	Approval before arrival	No queue at the gate
Pass	Handwritten slip	Unique pass number sent by e-mail	Traceable pass
Entry and exit	Manual times, often missing	Logged against the pass with status	Know who is inside
Reports	Tallied by hand	Reports and charts	Easier review

8.4 Limitations of the Low-Code Approach

The platform made the forms, approval process, reports and charts quick to build, but some limits were met. Features such as camera-based scanning, number-plate recognition and gate-hardware control are outside the platform’s visual builders and would need integration. The trial edition limits the period of use, and the platform dictates the available components and chart options. Logic that the builders cannot express, such as status updates and expiry, required Deluge scripts that had to be debugged through log messages. Workarounds include integration through APIs and a move to a paid edition for continued use.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

The application sits mostly at the low-code point of the spectrum: the forms, approval, reports and dashboard were configured visually, and short Deluge scripts supplied the logic. Deluge extends the no-code builders without leaving the platform, which is why most of the project could be completed with limited programming. The experience also showed that security and role settings still need technical understanding even on a low-code platform.

CHAPTER 9

CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Visitor Pass & Gate Entry Management System provides a structured form for requesting visitor passes, a host approval process that issues pass numbers, a gate log that records entry and exit, reports for reviewing visits and a dashboard of charts. Workflows and Deluge scripts automate notification, status changes and expiry. The project demonstrates that a useful visitor-management application can be built quickly on a low-code platform, and it identifies concrete steps for improvement. [Add one or two sentences on the observed test outcome.]

9.2 Future Enhancements

- Scan a QR code on the e-mailed pass at the gate instead of typing the pass number.
- Send SMS or WhatsApp notifications to the host and the visitor.
- Pre-approved recurring passes for vendors and contractors.
- Mobile app for guards and a portal for visitors to request passes.
- Blacklist of visitors and alerts for overstay beyond a set time.
- Integration with Zoho Analytics for advanced reports, and extended testing for invalid data, Production and user acceptance.

9.3 Learning Outcomes and Career Relevance

The project gave practical experience of building forms with lookups, an approval process, list reports, charts, a dashboard, form and scheduled workflows on Zoho Creator, of writing Deluge scripts, of reading the log to debug scripts, and of working with Stage and Production environments. The experience is relevant to roles such as low-code developer, business-application analyst and workflow consultant, in which quick delivery of business applications is valued [1], [2].

REFERENCES

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Left menu sections: Visitor Requests, Gate Desk, Masters, Analytics	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	Name, Phone, Email, Drop Down, Image, Lookup, Date, Time, Number	5.2	Tables 4.2–4.4
II	Lookup field (relationship between forms)	Y	Visitor_Pass_Request → Host_Employee (Host); Gate_Entry_Log → Visitor_Pass_Request (Pass)	4.2, 5.2	Fig. 4.2
II	Subform (parent–child data)	N	Not required by the data model	4.2	–
II	Field validations / layout rules	Y	DS-01, DS-02, mandatory fields	5.2, 5.5	Code C.1–C.2
II	List report with filter, grouping, custom action	Y	Pending Approvals (filter); Today's Visitors (group, Check-out button)	5.3	Table 5.1
II	Kanban / Calendar / Timeline / Map report	N	Not used	5.3	–
II	Pivot table / chart report	Y	Pie, bar and line chart views	5.3	Fig. 5.6
II	Page / dashboard built with UI Builder	[Y/N]	Visitor Pass Dashboard	5.4	Fig. 5.5
II	Deluge – form events (validate / success / field actions)	Y	DS-01, DS-02 On Validate; DS-03 On User Input; DS-04 On Success	5.5	Code C.1–C.4
II	Deluge – functions, fetch, loops, collections	Y	fetch in DS-03, DS-04; for each in DS-05	5.5	Code C.3–C.5
II	Email / in-app notifications	Y	Send-mail actions to host and visitor	5.6	Table 5.3
II	Form workflow	Y	New request; gate entry update	5.6	Table 5.3
II	Scheduled workflow	Y	Nightly expiry (DS-05)	5.6	Code C.5
II	Approval process	Y	Host approval	5.6	Fig. 5.8
II	Blueprint	N	Not used	5.6	–
II	Batch / report workflow	N	Not used	5.6	–
II	Testing & debugging evidence	[Y/N]	Test cases and defect log	6	Tables 6.1, 6.2
III	Integration with a Zoho service	Y	Zoho Mail send-mail (Zoho Cliq optional)	5.7	Table 5.4
III	Connection / REST API call (invokeurl)	[update]	–	5.7	–
III	Webhook or Creator REST API	[update]	–	5.7	–
III	Users, roles and permission profiles	Y	Four roles	5.8	Fig. 5.9
III	Record-level access criteria / customer portal	Y	Host sees own visitors	5.8	Table 4.5
III	Zia / AI feature	[update]	–	5.9	–
III	Analytics (charts / Zoho Analytics)	Y	Charts and dashboard	5.9	Fig. 5.5

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
IV	Performance optimisation applied & measured	[Y/N]	Table 7.2	7.2	Table 7.2
IV	Deployment across environments	[Y/N]	Stage and Production	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	Y	Permission profiles, record-level criteria	7.3	Table 7.3
IV	Governance (audit trail, administration)	[update]	–	7.4	–
V	Pro-code extension (Catalyst / widgets / custom functions)	[update]	–	5.10	–
V	Low-code limitations & spectrum analysis	Y	Sections 8.4 and 8.5	8.4, 8.5	–
—	Mobile app / portal access	[update]	–	5.11	–

Total features implemented: [update after final review].

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2		
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6		
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9		
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7		
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3		

APPENDIX C – COMPLETE DELUGE SOURCE CODE

The five Deluge scripts are listed in full. The form and field link names (Visit_Date, Mobile_Number, Host, Status and so on) must match those in the application.

C.1 Visitor_Pass_Request → On Validate (DS-01) – visit date not in the past

```
// Script ID : DS-01
if(input.Visit_Date < zoho.currentdate)
{
    alert "Visit date cannot be earlier than today";
    cancel submit;
}
```

C.2 Visitor_Pass_Request → On Validate (DS-02) – mobile number length

```
// Script ID : DS-02
digits = input.Mobile_Number.replaceAll("[^0-9]","");
if(digits.length() != 10)
{
    alert "Enter a valid 10-digit mobile number";
    cancel submit;
}
```

C.3 Host field → On User Input (DS-03) – fill department

```
// Script ID : DS-03
if(input.Host != null)
{
    hostRec = Host_Employee[ID == input.Host];
    input.Department = hostRec.Department;
}
```

C.4 Gate_Entry_Log → On Success (DS-04) – update pass status

```
// Script ID : DS-04
passRec = Visitor_Pass_Request[ID == input.Pass];
if(input.Exit_Time == null)
{
    passRec.Status = "Checked In";
    info "Pass " + passRec.Pass_Number + " checked in";
}
else
{
    passRec.Status = "Checked Out";
    info "Pass " + passRec.Pass_Number + " checked out";
}
```

C.5 Schedule, daily 11:59 PM (DS-05) – expire unused passes

```
// Script ID : DS-05
stale = Visitor_Pass_Request[Status == "Approved" && Visit_Date < zoho.currentdate];
count = 0;
for each rec in stale
{
    rec.Status = "Expired";
    count = count + 1;
}
info "Passes expired: " + count;
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

- Open the application URL given in the Project at a Glance in Google Chrome or Microsoft Edge.
- Sign in with a Zoho account that has been given access to the application.
- The application opens with the left-hand menu described in Section 5.1; the sections listed depend on your role.

D.2 Instructions for the Visitor / Receptionist

- Open Visitor Requests → Visitor Pass Request.
- Fill in the fields, starting with the mandatory fields marked with an asterisk, and select Submit. A confirmation message appears.
- Wait for the e-mail containing the approved pass number, and show it at the gate.

D.3 Instructions for the Host Employee

- Open the approval e-mail or Visitor Requests → Pending Approvals.
- Review the visitor details and select Approve or Reject.

D.4 Instructions for the Security Guard

- Open Gate Desk → Gate Entry Log and add a record: select the pass number, the gate and the entry time.
- When the visitor leaves, open Today’s Visitors and use Check-out to record the exit time.

D.5 Instructions for the Administrator

- Maintain hosts under Masters → Host Employees and users under Users.
- Open Analytics → Visitor Pass Dashboard to review the charts.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	PAVITHRAN	**Register No.**	241401065
Project Title	VISITOR PASS & GATE ENTRY MANAGEMENT SYSTEM	**Date**	03/11/26

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner
Name: _____

Signature of External Examiner
Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	PAVITHRAN	**Register No.**	241401065
Department / Section	B.Tech. COMPUTER SCIENCE AND BUSINESS SYSTEM	**Date of Viva**	03/11/26
Application URL Verified		**Demo Duration**	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER
Name: _____

EXTERNAL EXAMINER
Name: _____